

2024

Enhancing Phishing Attack Detection Using Explainable AI: Trends and Innovations

Bhagyashree D. Shendkar

Department of Computer Science and Engineering, MIT School of Computing, MIT Art Design and Technology University, Loni, Pune, India,, bhagyashree.d.shendkar@gmail.com

Pankaj R. Chandre

Department of Computer Science and Engineering, MIT School of Computing, MIT Art Design and Technology University, Loni, Pune, India

Sulochana Sagar Madachane

SIES Graduate School of Technology, Nerul, Navi Mumbai

Nikita Kulkarni

Department of Computer Engineering, K J College of Engineering and Management Research, Pune, India,

Sayalee Deshmukh

Department of Computer Science and, Engineering (AI & ML), Pimpri Chinchwad College of Engineering, Akurdi, Pune, India.

Follow this and additional works at: <https://ajstd.ubd.edu.bn/journal>



Part of the [Biotechnology Commons](#), [Engineering Commons](#), and the [Physical Sciences and Mathematics Commons](#)

Recommended Citation

Shendkar, Bhagyashree D.; Chandre, Pankaj R.; Madachane, Sulochana Sagar; Kulkarni, Nikita; and Deshmukh, Sayalee (2024) "Enhancing Phishing Attack Detection Using Explainable AI: Trends and Innovations," *ASEAN Journal on Science and Technology for Development*. Vol. 42: No. 1, Article 8.

DOI: <https://doi.org/10.61931/2224-9028.1604>

Available at: <https://ajstd.ubd.edu.bn/journal/vol42/iss1/8>

This Article is brought to you for free and open access by ASEAN Journal on Science and Technology for Development. It has been accepted for inclusion in ASEAN Journal on Science and Technology for Development by an authorized editor of ASEAN Journal on Science and Technology for Development.

ARTICLE

Enhancing Phishing Attack Detection Using Explainable AI: Trends and Innovations

Bhagyashree D. Shendkar^{a,*}, Sulochana S. Madachane^b, Pankaj R. Chandre^a,
Nikita Kulkarni^c, Sayalee Deshmukh^d

^a Department of Computer Science and Engineering, MIT School of Computing, MIT Art Design and Technology University, Loni, Pune, India

^b SIES Graduate School of Technology, Nerul, Navi Mumbai, India

^c Department of Computer Engineering, K J College of Engineering and Management Research, Pune, India

^d Department of Computer Science and, Engineering (AI & ML), Pimpri Chinchwad College of Engineering, Akurdi, Pune, India

Abstract

In the context of escalating cyber threats, phishing attacks continue to pose significant challenges to cybersecurity measures, exploiting human vulnerabilities to gain unauthorized access to sensitive information. Innovative solutions are required because traditional detection methods frequently fail to identify and mitigate these assaults successfully. To improve phishing attack detection, this study makes use of Explainable Artificial Intelligence (AI) tools, which provide decision-making processes with transparency and prediction accuracy. The paper begins by outlining the pervasive nature of phishing attacks in today's digital landscape, emphasizing their detrimental impact on individuals, organizations, and society at large. It then discusses the limitations of existing detection methods, which often lack interpretability and fail to provide insights into the underlying rationale behind their decisions. To tackle these challenges, the paper proposes the adoption of Explainable AI, which combines sophisticated machine learning algorithms with transparent decision-making mechanisms. The study examines the use of Explainable AI in phishing attack detection across a range of disciplines, including industry applications, contributions to academic research, and workable implementation tactics, through a survey of current trends and advances in the field. By elucidating the underlying principles of Explainable AI and its role in bolstering cybersecurity measures, this paper aims to provide valuable insights for researchers, practitioners, and policy-makers alike. In conclusion, the integration of Explainable AI holds immense promise for enhancing phishing attack detection capabilities, offering a balanced approach that prioritizes both accuracy and interpretability. By fostering transparency in decision-making processes and enabling human supervision and intervention, Explainable AI emerges as a pivotal tool in the ongoing battle against cyber threats, safeguarding digital assets and preserving trust in the digital ecosystem.

Keywords: Phishing attacks, Cybersecurity, Explainable Artificial Intelligence (AI), Detection methods, Transparency, Decision-making processes

1. Introduction

In the introduction section, the paper sets the stage by highlighting the pervasive threat of phishing

attacks in today's digital landscape. It gives a general overview of the difficulties that conventional detection approaches face due to the growing sophistication of phishing attempts (Sawe et al., 2024).



Received 24 August 2024; revised 17 October 2024; accepted 5 November 2024.
Available online 17 February 2025

* Corresponding author at: Department of Computer Science and Engineering, MIT Art, Design and Technology University, Loni Kalbhor, Pune, India.

E-mail addresses: bhagyashree.shendkar@mituniversity.edu.in, bhagyashree.d.shendkar@gmail.com (B.D. Shendkar), sulochana.madachane@gmail.com (S.S. Madachane), pankaj.chandre@mituniversity.edu.in, pankajchandre30@gmail.com (P.R. Chandre), nikitakulkarni.kjcoemr@kjei.edu.in (N. Kulkarni), sayalee.deshmukh@pccoepune.org (S. Deshmukh).

Moreover, it presents Explainable AI (XAI) as a promising method to improve phishing detection by bringing interpretability and transparency to detection models' decision-making process (Molay, 2009). In addition to providing background information for examining current developments and advancements in phishing attack detection, the section seeks to highlight the significance of implementing XAI approaches in cybersecurity.

1.1. Overview of phishing attacks

Cybercriminals use phishing assaults, which are deceptive techniques, to trick people into disclosing private information like credit card numbers, passwords, or personal identification numbers (Khonji et al., 2013). These assaults frequently entail pretending to be reputable organisations, like banks, social media sites, or government organisations, and deceiving people into opening dangerous links or divulging personal information via phoney emails, texts, or websites (Williams & Joinson, 2020). Phishing attacks are a major worry for people, businesses, and organisations around the world since they can have serious repercussions such as identity theft, financial losses, and data breaches.

1.2. Importance of phishing attack detection

To lessen their effects and guard against possible harm, phishing attack detection is essential. Individuals and organisations can take proactive steps to prevent unauthorised access to sensitive information and minimise potential damage by promptly identifying phishing efforts (Sutter et al., 2022). Strong phishing detection systems contribute to the protection of user confidence, the security of financial and personal information, and the upkeep of digital networks and systems (Liu et al., 2018). The development of strong detection systems is crucial for improving cybersecurity posture and lowering the likelihood that successful phishing attempts will occur, given the growing sophistication and frequency of phishing attacks.

1.3. Introduction to explainable AI in phishing detection

Explainable Artificial Intelligence (XAI) refers to the transparency and interpretability of machine learning models and algorithms, allowing users to understand the rationale behind their decisions (Demertzis & Iliadis, 2019). Cybersecurity experts may evaluate the dependability of detection methods and take appropriate action when needed

by using XAI approaches, which offer insights into how detection systems evaluate and categorise possible phishing attacks. Organisations can improve the reliability and efficacy of their cybersecurity measures, facilitating improved decision-making and reaction to dynamic threats, by integrating XAI into phishing detection systems.

2. Background and related work

2.1. Historical perspective of phishing attacks

Phishing attacks have been a persistent threat since the early days of the internet. The phrase “phishing” comes from the analogy of “fishing,” in which cybercriminals utilise bait to trick victims into divulging private information like passwords, usernames, and bank account information (Punia et al., 2021). Phishing attempts were first straightforward, frequently using emails that fooled people into believing they were from reputable companies. But phishing techniques also become more sophisticated as technology advanced. Attackers started employing a variety of strategies, including spear phishing, social engineering, and whaling, to target particular people or organisations (Butt et al., 2023). Phishing attacks have grown more common over time and are still evolving, which presents serious difficulties for cybersecurity.

2.2. Existing techniques for phishing attack detection

To combat the growing threat of phishing attacks, researchers and cybersecurity experts have developed various techniques for detection and prevention. Conventional methods involve rule-based systems that detect phishing attempts by using pre-established patterns or signatures (Sanchez-Panagua et al., 2022). These systems are somewhat effective, but they frequently find it difficult to stay up with attackers' constantly changing strategies. More sophisticated methods use machine learning algorithms to examine user behaviour, URLs, and email content in order to identify questionable activity (Orunsolu et al., 2022). Through the process of learning from fresh data, these methods can offer increased precision and flexibility. However, it could be difficult to comprehend how they arrive at their conclusions because of their lack of interpretability and transparency.

The study (Chinnasamy et al., 2024) discusses the critical role of 6G-enabled Smart Grid-based Cyber-Physical Systems (SGCPS) in ensuring secure and efficient energy infrastructure amid technological

advancements. It emphasizes the need for Intelligent Breach Detection Systems (IBDS) to mitigate increased cyber threats stemming from the vast connectivity offered by 6G. The paper reviews key methodologies for intrusion detection, including AI-based systems, anomaly detection, and robust state estimators. A novel model, AI-IBDS, is introduced, leveraging the Grey Wolf Algorithm and Artificial Neural Networks (GWAANN) for optimized intrusion detection in energy systems. Comparative analysis shows the GWAANN outperforms traditional methods like SVM and KNN, enhancing network security and resilience in SGCPs environments.

The study (Chinnasamy et al., 2023) provides an overview of deep learning, starting with its historical development and current global landscape. It explains the fundamental structure and characteristics of deep learning, covering key techniques such as stacked autoencoders, deep belief networks, deep Boltzmann machines, and convolutional neural networks. The study also explores recent advancements and applications in fields like speech recognition, machine learning, computational linguistics, and healthcare. Finally, it addresses the challenges and future opportunities for deep learning research, highlighting areas for continued growth and innovation.

The study (Chinnasamy et al., 2022) provides phishing poses a significant risk to personal information, targeting individuals, cloud storage providers, and government agencies. While existing anti-phishing methods, such as hardware-based solutions, are not cost-effective, software-based techniques are more commonly used. However, current models struggle to address the zero-day phishing problem. To tackle this, the study proposes a heuristic approach for phishing detection, classifying URLs as phishing or non-phishing based on features like web traffic and URLs. The methodology employs machine learning algorithms such as Random Forest, SVM, and Genetic Algorithms, using phishing datasets to improve detection accuracy.

The study (Chinnasamy et al., 2023) proposes a Distributed Denial of Service (DDoS) attack involves multiple compromised computers, known as “zombies,” being used to target and overwhelm a service. Attackers exploit software vulnerabilities and use malicious agents on these computers to launch attacks. As DDoS tactics evolve, it becomes crucial to develop defenses that address both current and future variations. This research discusses DDoS attack types, terminology, detection systems, and countermeasures, with a focus on defending cloud computing environments. Understanding the scope of DDoS attacks enables the creation of more effective protective strategies for future threats.

The study (Chinnasamy et al., 2024) proposes an advanced system leveraging artificial intelligence (AI) to proactively prevent and detect phishing attacks, which pose a significant cybersecurity threat by exploiting human vulnerabilities to steal sensitive information. It employs machine learning algorithms and natural language processing (NLP) to analyze email, website, and digital interaction characteristics to accurately identify phishing attempts. By integrating AI-driven methods, the system enhances existing cybersecurity defenses, providing a more reliable and efficient approach to protecting individuals and organizations from the harmful effects of phishing attacks.

Table 1 outlines various existing techniques for phishing attack detection, each with its distinct advantages and limitations. Rule-based systems can be susceptible to new attacks and false positives, despite their ease of use and speedy recognition of well-known patterns. While machine learning techniques are excellent at identifying changing strategies, they may be difficult to interpret and require labelled data. While text and URL analysis concentrates on linguistic and structural irregularities, it may overlook sophisticated attacks or produce false positives. Real-time detection is possible with behavioural analysis, but it also presents privacy issues and may produce false positives. Although sender reputation analysis and content screening provide customisable filtering, skilled attackers may be able to circumvent them or reject valid communications. The Table 2 shows the summary of explainable AI techniques. All things considered; a blend of these methods adapted to organisational requirements is essential for successful phishing detection.

Fig. 1 shows the various types of phishing attacks, all of which aim to deceive users and extract sensitive information. At the core, “Phishing Attacks” represents the umbrella term under which different phishing strategies fall. These attacks can take multiple forms. One of the most prevalent is Email Phishing, where attackers send fraudulent emails posing as legitimate entities. Spear Phishing goes a step further by targeting specific individuals or organizations with personalized content, making it harder to detect. Similarly, Whaling focuses on high-profile targets like executives, often using highly sophisticated techniques. Vishing, on the other hand, uses voice calls to trick users into revealing confidential information, while Smishing relies on deceptive SMS messages to achieve the same. Despite differing in their delivery methods, all these attacks share the common objective of manipulating users to give up personal data or access credentials.

Table 1. Summary of existing techniques for phishing attack detection.

Technique	Description	Advantages	Limitations
Rule-based systems	Utilize predefined rules or patterns to identify phishing attempts. These rules may include specific keywords, URL patterns, or email attributes.	- Simple and easy to implement - Can quickly identify known phishing patterns	- Limited effectiveness against novel or sophisticated phishing attacks - Prone to false positives and negatives due to rigid rule sets
Machine learning	Employ various machine learning algorithms to analyze email content, URLs, and user behavior for detecting suspicious activities.	- Capable of detecting unknown or evolving phishing tactics - Can handle large volumes of data and identify complex patterns	- Requires labeled training data for model training - Susceptible to adversarial attacks and model biases - Interpretability may be limited depending on the algorithm
Text analysis	Analyze the textual content of emails, focusing on linguistic features, grammar, spelling errors, and stylistic inconsistencies.	- Able to detect phishing attempts based on language cues and context - Effective against phishing emails with social engineering tactics	- May produce false positives due to variations in language and writing styles - Limited to text-based phishing attacks
URL analysis	Scrutinize URLs embedded in emails or web pages for anomalies such as misspellings, redirects, or suspicious domain names.	- Identifies malicious URLs and prevents users from accessing phishing sites - Relatively fast and scalable method for detecting phishing links	- May miss sophisticated phishing URLs that closely mimic legitimate websites - Cannot detect phishing attempts without URLs
Behavioral analysis	Monitor user behavior, such as click patterns, response times, and interaction history, to identify deviations indicative of phishing attempts.	- Detects anomalies in user behavior that may indicate phishing activity - Provides real-time detection and response to phishing attempts	- Requires access to user behavior data and may raise privacy concerns - False positives may occur due to legitimate variations in user behavior
Sender reputation analysis	Assess the reputation of email senders or domains based on past behavior, blacklists, or whitelists maintained by cybersecurity organizations.	- Identifies emails from known malicious senders and prevents them from reaching users - Can block emails from domains with poor reputation scores	- May block legitimate emails from new or lesser-known senders - Effectiveness depends on the accuracy and currency of reputation data
Content filtering	Apply filters to emails based on predefined criteria, such as sender address, subject line, attachments, or embedded links.	- Enables organizations to customize filtering rules based on their security policies - Helps prevent users from interacting with potentially harmful content	- May generate false positives if filtering criteria are too strict or too broad - Can be bypassed by sophisticated attackers through social engineering tactics

Table 2. Summary of explainable AI techniques.

Explainable AI Techniques	Description	Advantages	Limitations
LIME (Local Interpretable Model-agnostic Explanations)	Generates local explanations for model predictions by perturbing input features	Provides insight into model behavior at the instance level, helps understand model decisions	Limited to explaining individual pre-dictions, may not capture global model behavior
SHAP (SHapley Additive exPlanations)	Assigns a value to each feature to explain its contribution to the model prediction	Offers global explanations for model predictions, captures feature interactions	Computationally intensive for large datasets, may be challenging to interpret for complex models
Decision trees	Represent models as a series of if-else decision rules	Easy to understand and interpret, provides transparent decision-making process	Prone to overfitting, may not capture complex relationships in data
Partial dependence plots	Visualizes the effect of a single feature on the model prediction while marginalizing the effects of other features	Offers intuitive understanding of individual feature effects, helps identify non-linear relationships	Limited to analyzing one feature at a time, may not capture feature interactions
LRP (Layer-wise Relevance Propagation)	Propagates relevance scores backward through the neural network to attribute importance to input features	Provides insight into how each input feature contributes to the model output	Limited interpretability for deep neural networks, may not capture complex interactions in hidden layers
Integrated gradients	Computes feature attributions by integrating the gradients of the model prediction with respect to input features	Offers intuitive explanations for model predictions, captures feature importance across the input space	Computationally expensive for large models and high-dimensional data, requires access to model gradients

2.3. Review of explainable AI in cybersecurity

A potential remedy for the interpretability and transparency problems with machine learning-based phishing detection systems is explainable AI (XAI). The goal of XAI approaches is to give cybersecurity analysts more understanding and confidence in the outputs of AI models by offering insights into the decision-making process of these algorithms (Jawale et al., 2020; Sánchez-Paniagua et al., 2022). When it comes to phishing detection, XAI techniques can assist in determining which characteristics or aspects are most important in classifying an email or URL as real or phishing. XAI helps analysts more efficiently evaluate false positives or negatives and confirm the veracity of detection results by offering comprehensible explanations (van Geest et al., 2024). All things considered, XAI has enormous potential to improve the reliability and efficacy of phishing detection systems in cybersecurity.

The examination of explainable AI methods in cybersecurity emphasises how important they are for giving complicated machine learning models used to detect phishing attacks clarity and interpretability. Different approaches provide different benefits, such as local interpretability with LIME, global explanations with SHAP, and decision trees and partial dependency plots that facilitate intuitive understanding. These techniques do have several drawbacks, though, including computational cost, difficulties with deep neural network interpretability, and limitations in capturing intricate interactions. Notwithstanding these difficulties, the incorporation of explainable AI approaches continues to be essential for enhancing the reliability and efficiency of phishing attack detection systems, allowing stakeholders to have a better understanding of and ability to counteract such risks.

3. Recent trends in phishing attack detection

Recent trends in phishing attack detection have seen a significant focus on the utilization of Explainable AI (XAI) models. Because of their openness and interpretability, these models help cybersecurity professionals comprehend the thought process that goes into identifying phishing efforts (Ribeiro et al., 2024). Using methods like SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-Agnostic Explanations), XAI models enhance the reliability of detection systems by offering insights into the qualities that are most important in recognising phishing assaults. The direct incorporation of explainable elements

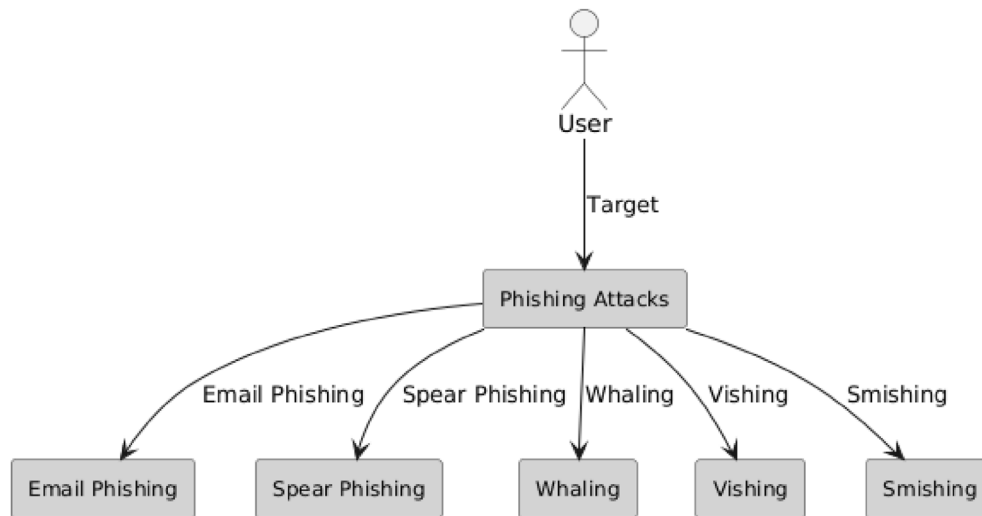


Fig. 1. Various types of phishing attacks

into phishing detection systems is another new trend. Researchers studying cybersecurity are integrating interpretable characteristics into their algorithms rather than depending only on black-box models (Safi & Singh, 2023). These capabilities, which include sender reputation scores, content-based heuristics, and URL structure analysis, not only make the system easier to use but also increase its overall accuracy and resistance to phishing attempts that are always evolving. Furthermore, case studies and actual situations are providing evidence for current advancements in the detection of phishing attacks. These studies offer insightful information about the efficacy of various detection techniques, the difficulties associated with real-world implementations, and the possible effects of phishing assaults on people and organisations. Through the examination of historical events and contemporary patterns, scientists can enhance their methods of identification and create stronger barriers against phishing attacks.

3.1. Utilization of explainable AI models

Explainable AI (XAI) models have gained prominence in recent years due to their ability to provide insights into the decision-making process of complex machine learning algorithms. Researchers are using XAI models more frequently in the context of phishing attack detection to improve interpretability and transparency (Opara et al., 2024). These models aid analysts and cybersecurity specialists in comprehending how the detection system arrived at a specific conclusion or prediction (Chandre et al., 2018). Phishing detection systems are using techniques like SHAP (SHapley Additive exPlanations) and LIME

(Local Interpretable Model-agnostic Explanations) to provide global and localised explanations, respectively (Palaniappan et al., 2020). Researchers hope to give users meaningful insights into potential risks and enhance the reliability of phishing detection systems by employing XAI models.

3.2. Integration of explainable features in phishing detection systems

Another trend in phishing attack detection involves the integration of explainable features directly into detection systems. Explainable features are qualities that make phishing emails, URLs, or online material easy for people to understand and interpret (Remmide et al., 2022). These characteristics aid in differentiating between trustworthy and harmful organisations and act as warning signs of possible phishing efforts. Explainable aspects include things like the length of the URL, misspellings, HTTPS usage, and domain reputation. Researchers hope to increase the precision and comprehensibility of detection systems by adding these characteristics to phishing detection algorithms, empowering consumers to decide for themselves whether an online piece of content is legitimate.

This Fig. 2 represents the components involved in enhancing phishing attack detection using Explainable AI. It consists of three components: the Explainable AI System (XAI Model, Interpretation Module), the Network (Firewall, Web Server, Email Server), and the Detection System (Phishing Detection Module). The links between these parts show how the Phishing Detection Module processes email traffic from the Email Server and HTTP traffic from the Web Server after it has been watched by the Firewall. The Explainable AI System receives the

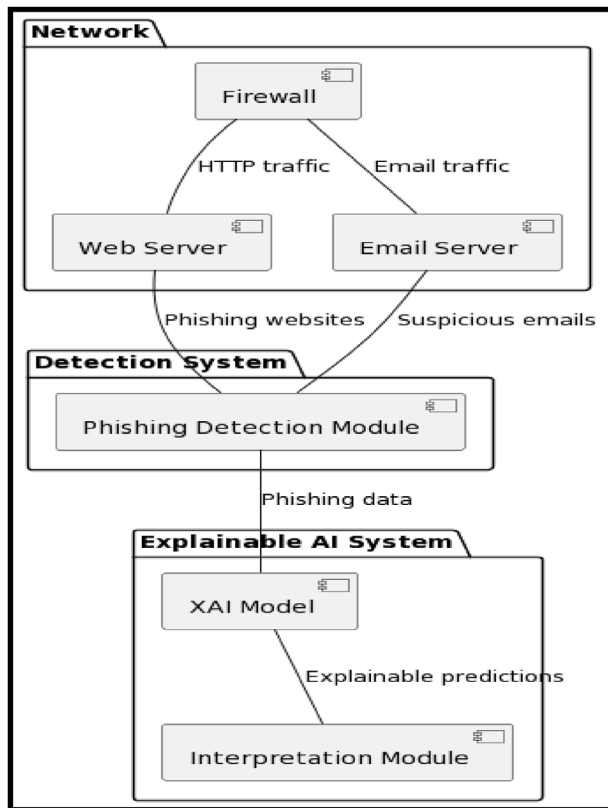


Fig. 2. An architecture Enhancing Phishing Attack Detection Using Explainable AI

detection process results and interprets them to produce explainable forecasts that support decision-making.

3.2.1. Network components

3.2.1.1. Firewall. Acts as a barrier between the internal network (Web Server, Email Server) and external sources. It monitors incoming and outgoing traffic, including HTTP traffic from the Web Server and email traffic from the Email Server.

3.2.1.2. Web server. Hosts web applications and websites. It can be a target for phishing attacks, as malicious actors may attempt to host deceptive websites to steal user credentials.

3.2.1.3. Email server. Handles incoming and outgoing email messages. Phishing attacks often involve sending deceptive emails containing links to fraudulent websites or attachments containing malware.

3.2.2. Explainable AI system components

3.2.2.1. XAI model. Represents the machine learning model used for phishing attack detection. It is designed to provide not only accurate predictions

but also explanations for why certain decisions were made.

3.2.2.2. Interpretation module. Responsible for interpreting the predictions generated by the XAI Model. It contributes to the transparency and comprehensibility of the decision-making process by illuminating the model's inner workings.

3.2.3. Detection system component

3.2.3.1. Phishing detection module. Examines incoming data to find possible phishing attacks, such as HTTP requests or email content. It uses machine learning methods to identify patterns suggestive of phishing activity, potentially involving Explainable AI.

3.2.3.2. Connections. The Firewall is used to monitor HTTP traffic from the Web Server and email traffic from the Email Server.

The Phishing Detection Module receives suspicious emails and website requests that the Firewall has detected and forwards them for examination.

After processing the data, the Phishing Detection Module forwards it to the XAI Model for forecasting.

The Interpretation Module interprets the predictions produced by the XAI Model to give reasons for the choices that were taken.

In general, this architecture enhances the detection of phishing attacks while offering transparency and interpretability in the decision-making process. It does this by combining network security measures (Firewall) with machine learning-based phishing detection (Phishing Detection Module) and Explainable AI techniques (XAI Model and Interpretation Module).

The major contribution of this proposed work is the development of a phishing detection system integrated with explainable artificial intelligence (XAI) to enhance both detection accuracy and transparency. By combining a phishing detection module with an XAI model, the system identifies phishing websites and suspicious emails while providing human-understandable explanations for its predictions. This integration improves trust and interpretability in security-critical environments. Additionally, the system monitors both HTTP and email traffic through firewalls, covering multiple attack vectors, which strengthens its overall threat detection capabilities and makes it a comprehensive solution for phishing detection.

3.3. Case studies and examples

Analysing real-world case studies and examples to better understand emerging phishing methods

and strategies is one of the latest trends in phishing attack detection research. Scholars scrutinise past phishing efforts, investigate attack trends, and assess how well-suited current detection methods are for spotting complex assaults (Hr et al., 2020; Kore et al., 2015). Case studies offer important insights into the strategies employed by cybercriminals, the weaknesses they take advantage of, and the effects they have on both persons and organisations. Researchers can improve current detection techniques, create fresh tactics, and strengthen phishing detection systems' resistance to new threats by looking at real-world cases.

4. Challenges and limitations

4.1. Interpretability vs. accuracy trade-offs

In phishing attack detection using explainable AI, there often exists a trade-off between interpretability and accuracy. Complex machine learning models can be more accurate in spotting phishing attempts, but they can also be harder to read, which makes it harder to comprehend the underlying reasoning behind the model. Conversely, less complex models or those with inherent interpretability features might forgo some accuracy in exchange for clearer forecasts and justifications (Paliath et al., 2020). Depending on the needs of their application, researchers and practitioners must carefully weigh these trade-offs. Better comprehension and confidence in the detection system are made possible by approaches like SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-agnostic Explanations), which provide methods to improve interpretability without materially sacrificing accuracy.

4.2. Scalability issues

Another significant challenge in phishing attack detection using explainable AI is scalability. The intricacy of phishing attempts and the amount of data they include could make it difficult for typical machine learning models to scale effectively (Damre et al., 2024; Das Gupta et al., 2024). Deploying such systems in real-time applications or devices with limited resources may become impractical due to the potential for lengthier processing times and additional computational requirements. The creation of innovative methods and algorithms that can manage big datasets and intricate feature spaces without compromising performance is necessary to address scalability concerns (Kathrine et al., 2019). Furthermore, by enabling parallelized processing and resource provisioning as needed, cloud-based

solutions and distributed computing frameworks can assist reduce scaling issues.

4.3. Ethical considerations

Ethical considerations play a crucial role in the development and deployment of phishing attack detection systems using explainable AI. Ensuring user privacy, fairness, and transparency is crucial since these systems may analyse and act upon sensitive user data (Gupta et al., 2021). Concerns like algorithmic bias and fairness, data privacy and consent, and decision-making accountability and openness are examples of ethical considerations. To protect user rights and minimise any harm, researchers and practitioners need to abide by ethical principles and regulatory frameworks. Additionally, while resolving ethical concerns, incorporating stakeholders including end users in the development process and giving them comprehensive explanations of how the system works can promote confidence and adoption of the technology.

5. Future directions

5.1. Advancements in explainable AI techniques

Explainable AI (XAI) techniques have seen significant advancements in recent years, contributing to improved transparency and interpretability of machine learning models used for phishing attack detection (Bhujbal et al., 2024). Innovative techniques for comprehending and interpreting the decision-making process of complicated models include Integrated Gradients, SHapley Additive exPlanations (SHAP), and Local Interpretable Model-agnostic Explanations (LIME). By approximating the behaviour of the model around particular examples, LIME generates local explanations that facilitate the understanding of individual predictions. By measuring the significance of every feature in the model's output, SHAP improves overall transparency by offering global explanations (Salloum et al., 2022). By providing information on feature contributions over the whole input space, integrated gradients help to clarify the behaviour of the model. By providing cybersecurity professionals with important insights into the inner workings of phishing detection models, these developments in XAI approaches enhance trust and trustworthiness.

5.2. Hybrid approaches for enhanced detection

Hybrid approaches, which combine traditional rule-based techniques with machine learning

algorithms, have emerged as a promising strategy for enhancing phishing attack detection. Hybrid models can effectively overcome the limits of separate methods and increase overall detection accuracy by utilising the benefits of both approaches (Bojjagani et al., 2020; Makubhai et al., 2023). Rule-based systems, on the other hand, use predefined rules to instantly recognise known phishing patterns, while machine learning algorithms use previous data to adapt to new and changing threats. By combining these two strategies, more resilient and adaptable detection systems that can recognise known and unidentified phishing attacks can be created. Furthermore, hybrid techniques increase performance across various datasets and settings by providing more flexibility in feature selection and model optimisation.

5.3. Integration with real-time monitoring systems

For prompt threat detection and response, real-time monitoring systems and phishing attack detection systems must be integrated. Through the continuous observation of network traffic, email exchanges, and user interactions made possible by real-time monitoring, businesses may identify questionable activity and possible phishing attempts as soon as they happen (Sadiq et al., 2021). Security analysts can obtain a better understanding of risks that have been spotted, comprehend the reasoning behind model predictions, and make well-informed decisions regarding incident response tactics by including explainable AI approaches into these monitoring systems. Furthermore, by facilitating quick feedback loops, real-time monitoring systems help organisations refine their detection models iteratively based on feedback from the actual world and newly discovered threat intelligence. The integration in question guarantees the continued effectiveness and adaptability of phishing attack detection in changing cybersecurity contexts, hence augmenting organisational resilience against cyber-attacks.

5.4. Tool discussions

In enhancing phishing attack detection using Explainable AI (XAI), a comprehensive tools discussion is crucial for understanding the practical implementation of the proposed methods. Machine learning frameworks like TensorFlow, Scikit-learn, and PyTorch serve as the backbone for developing and training models that can identify phishing attempts. These frameworks offer a range of pre-built

algorithms that can be tailored to suit specific detection requirements. Additionally, the integration of XAI libraries such as SHAP (SHapley Additive exPlanations) and LIME (Local Interpretable Model-agnostic Explanations) allows for a deeper understanding of model decisions, providing transparency through visualizations that highlight the most influential features in predictions. This transparency is essential in building trust with stakeholders, enabling them to comprehend how the models operate and the rationale behind their decisions. Furthermore, the implementation of data preprocessing tools like Pandas, BeautifulSoup, and natural language processing libraries (NLTK or SpaCy) is vital for preparing the dataset and extracting relevant features from emails and web pages. The integration of threat intelligence platforms, such as Recorded Future and Anomali, can enrich the detection models with real-time insights into evolving phishing tactics, thereby improving accuracy. Visualization tools like Tableau and Power BI can present results effectively, allowing stakeholders to assess the performance of the detection system visually. Additionally, using deployment frameworks like Docker or Kubernetes ensures scalability and efficient integration within existing infrastructures. Continuous monitoring tools such as Grafana and Prometheus further enhance the detection system by enabling real-time performance tracking and proactive adjustments to maintain effectiveness. Together, these tools facilitate a robust and effective approach to phishing attack detection, significantly contributing to cybersecurity efforts.

6. Applications and use cases

6.1. Industry applications of explainable AI in phishing detection

Explainable AI (XAI) has gained significant traction in the industry, particularly in the realm of cybersecurity and phishing attack detection. Cybersecurity companies and organisations use XAI approaches in real-world applications to improve the interpretability and transparency of their phishing detection systems (Ahsan et al., 2022). For example, businesses use XAI models to give security analysts comprehensive justifications for why a specific email or website is considered potentially dangerous, so they may make well-informed decisions. Additionally, XAI assists in spotting patterns and trends in phishing assaults, enabling businesses to proactively bolster their security measures.

6.2. Academic research contributions

In academia, researchers have been actively exploring the application of explainable AI in phishing attack detection. Numerous research projects concentrate on creating cutting-edge XAI algorithms designed especially to detect phishing attempts with a high degree of interpretability and accuracy. The creation and assessment of machine learning models that not only identify phishing attempts but also offer thorough justifications for their conclusions is a common task for academic contributors. In addition, scientists investigate how well various XAI approaches work in practical settings and work to make phishing detection systems more resilient and scalable.

6.3. Practical implementations and deployment strategies

Practical implementations of XAI in phishing detection encompass a range of deployment strategies aimed at maximizing the effectiveness of cybersecurity measures. Businesses use XAI models in conjunction with online browsers, email filtering systems, and network security appliances as part of their current security setup (Maseno, 2017). The creation of user-friendly interfaces that enable non-technical staff to efficiently comprehend the explanations offered by XAI models may also be a component of deployment plans. Furthermore, to maintain maximum performance over time and respond to changing cyber threats, XAI-based phishing detection systems must be continuously monitored and improved.

The use case of enhancing phishing attack detection with Explainable AI (XAI) is particularly relevant for organizations in the financial sector, where the volume of incoming emails poses significant risks. Traditional phishing detection methods often rely on black-box machine learning models that, while achieving high accuracy, lack transparency. This opacity can hinder security analysts' trust in the predictions made by these models. To address this challenge, organizations can gather historical email data, including legitimate and phishing examples, to train a machine learning model using features such as email headers, sender reputation, and message content. By integrating XAI techniques, such as SHAP and LIME, the model can provide insights into its decision-making process. For each classified email, the system generates explainability reports detailing the key features influencing the classification—such as suspicious URL patterns or anomalous language usage—thus improving

analysts' understanding and trust in the model. This approach enables real-time detection of phishing attempts, allowing security teams to efficiently prioritize their investigations, focus on high-risk flagged emails, and reduce the time spent on false positives. Additionally, the insights gained can be utilized for employee education on recognizing phishing tactics, ultimately fostering a proactive cybersecurity culture. Overall, combining XAI with phishing detection enhances not only detection capabilities but also the effectiveness of incident response and user awareness initiatives.

7. Conclusion

In conclusion, this paper has explored the potential of Explainable Artificial Intelligence (AI) techniques in enhancing phishing attack detection, addressing critical gaps in existing cybersecurity measures. Through a comprehensive review of recent trends and innovations, several key findings have emerged. Firstly, phishing attacks remain a pervasive threat in the digital landscape, necessitating advanced detection methodologies that offer both accuracy and transparency. Secondly, existing detection methods often lack interpretability, hindering their effectiveness in providing actionable insights into malicious activities. These issues can be resolved with Explainable AI integration, which offers a well-rounded strategy that blends predictive accuracy with open decision-making procedures. Explainable AI improves cybersecurity systems' reliability and permits human oversight and intervention by clarifying the underlying reasoning behind detection decisions. Furthermore, the study has emphasised the value of Explainable AI in promoting accountability and transparency in phishing detection, hence enhancing resistance to changing cyberthreats. Looking ahead, several suggestions have been made for new research topics. These include developing standardised frameworks for assessing the interpretability and performance of Explainable AI models in real-world scenarios, exploring new Explainable AI techniques specifically suited for phishing attack detection, and researching the incorporation of human-centric design principles into AI-driven cybersecurity systems. **The cybersecurity community can keep improving the state-of-the-art in phishing detection and strengthen the resistance of digital ecosystems against new threats by focusing on these study areas.**

Ethics information

This research did not require ethical approval.

Funding

This research received no funding.

Declaration of AI Technology Usage

“We declare that the following Artificial Intelligence (AI) technologies and AI-assisted tools were employed in the preparation of this article:

Plantuml: Used for figure generation.

ChatGPT: Assisted in grammar checking, some content generation and summarization.

Conflict of interest

Author has no conflict of interest.

References

- Ahsan, M., Nygard, K. E., Gomes, R., Chowdhury, M. M., Rifat, N., & Connolly, J. F. (2022). Cybersecurity threats and their mitigation approaches using machine learning—a review. *Journal of Cybersecurity and Privacy*, 2(3), 527–555. <https://doi.org/10.3390/jcp2030027>
- Bhujbal, P. M., Jadhav, A., Nandimath, J. N., Kadam, P. S., Chandre, P. R., & Mahalle, P. N. (2024). *Intelligent systems and applications in engineering zero trust paradigm : Advancements , challenges , and future directions in cybersecurity*.
- Bojjagani, S., Brabin, D. R. D., & Rao, P. V. V. (2020). Phish-Preventer: A secure Authentication Protocol for prevention of phishing attacks in Mobile environment with formal verification. *Procedia Computer Science*, 171, 1110–1119. <https://doi.org/10.1016/j.procs.2020.04.119>, 2019.
- Butt, U. A., Amin, R., Aldabbas, H., Mohan, S., Alouffi, B., & Ahmadian, A. (2023). Cloud-based email phishing attack using machine and deep learning algorithm. *Complex and Intelligent Systems*, 9(3), 3043–3070. <https://doi.org/10.1007/s40747-022-00760-3>
- Chandre, P. R., Mahalle, P. N., & Shinde, G. R. (2018). Machine learning based novel approach for intrusion detection and prevention system: A tool based verification. In *2018 IEEE global conference on wireless computing and networking (GCWCN)* (pp. 135–140). <https://doi.org/10.1109/GCWCN.2018.8668618>
- Chinnasamy, P., Devika, S., Balaji, V., Dhanasekaran, S., Jebamani, B. J. A., & Kiran, A. (2023). BDDoS: Blocking distributed denial of Service flooding attacks with dynamic path detectors. In *2023 international conference on computer communication and informatics (ICCCI)*, Coimbatore, India (pp. 1–5). <https://doi.org/10.1109/ICCCI56745.2023.10128499>
- Chinnasamy, P., Krishnamoorthy, P., Alankruthi, K., Mohanraj, T., Kumar, B. S., & Chandran, L. (2024). AI enhanced phishing detection system. In *2024 third international conference on intelligent techniques in control, optimization and signal processing (INCOS)*, Krishnankoil, Virudhunagar district, Tamil Nadu, India (pp. 1–5). <https://doi.org/10.1109/INCOS59338.2024.10527485>
- Chinnasamy, P., Kumaresan, N., Selvaraj, R., Dhanasekaran, S., Ramprathap, K., & Boddu, S. (2022). An efficient phishing attack detection using machine learning algorithms. In *2022 international conference on advancements in Smart, secure and intelligent computing (ASSIC)* (pp. 1–6). Bhubaneswar, India. <https://doi.org/10.1109/ASSIC55218.2022.10088399>.
- Chinnasamy, P., Samrin, R., Sujitha, B. B., et al. (2024). Integrating intelligent Breach detection system into 6 g enabled smart grid-based cyber physical systems. *Wireless Personal Communications*. <https://doi.org/10.1007/s11277-024-11192-2>.
- Chinnasamy, P., Sathya, K. B., Jebamani, B. J., Nithyasri, A., & Fowjiya, S. (2023). Deep learning: Algorithms, techniques, and applications — a systematic survey. In L. Ashok Kumar, D. Karthika Renuka, & S. Geetha (Eds.), *Deep learning research applications for natural language processing* (pp. 1–17). IGI Global. <https://doi.org/10.4018/978-1-6684-6001-6.ch001>.
- Damre, S. S., Shendkar, B. D., Kulkarni, N., Chandre, P. R., & Deshmukh, S. (2024). Smart healthcare wearable device for early disease detection using machine learning. *International Journal of Intelligent Systems and Applications in Engineering*, 12(4s), 158–166.
- Das Gupta, S., Shahriar, K. T., Alqahtani, H., Als Salman, D., & Sarker, I. H. (2024). Modeling hybrid feature-based phishing websites detection using machine learning techniques. *Annals of Data Science*, 11(1), 217–242. <https://doi.org/10.1007/s40745-022-00379-8>
- Demertzis, K., & Iliadis, L. (2019). Cognitive web application firewall to critical infrastructures protection from phishing attacks. *Journal of Computations & Modelling*, 9(2), 1792–8850.
- Gupta, B. B., Yadav, K., Razzak, I., Psannis, K., Castiglione, A., & Chang, X. (2021). A novel approach for phishing URLs detection using lexical based machine learning in a real-time environment. *Computer Communications*, 175(April), 47–57. <https://doi.org/10.1016/j.comcom.2021.04.023>
- Hr, M. G., Mv, A., Gunesh Prasad, S., & Vinay, S. (2020). Development of anti-phishing browser based on random forest and rule of extraction framework. *Cybersecurity*, 3(1), 1–14. <https://doi.org/10.1186/s42400-020-00059-1>
- Jawale, A., Warole, P., Bhandare, S., Bhat, K., & Chandre, R. (2020). Jeevn-Net: Brain tumor segmentation using cascaded U-net & overall survival prediction. *International Research Journal of Engineering and Technology*, 56–62.
- Kathrine, G. J. W., Praise, P. M., Rose, A. A., & Kalaivani, E. C. (2019). Variants of phishing attacks and their detection techniques. In *Proceedings of the international conference on trends in electronics and informatics, ICOEI 2019, Icoei* (pp. 255–259). <https://doi.org/10.1109/ICOEI.2019.8862697>
- Khonji, M., Iraqi, Y., & Jones, A. (2013). Phishing detection: A literature survey. *IEEE Communications Surveys and Tutorials*, 15(4), 2091–2121. <https://doi.org/10.1109/SURV.2013.032213.00009>
- Kore, V. S., Tidke, B. A., & Chandre, P. (2015). Survey of image retrieval techniques and algorithms for image-rich information networks. *International Journal of Computer Application*, 112(6), 39–42. <https://www.ijcaonline.org/archives/volume112/number6/19674-1244%0Ahttp://research.ijcaonline.org/volume112/number6/pxc3901244.pdf>.
- Liu, L., De Vel, O., Han, Q. L., Zhang, J., & Xiang, Y. (2018). Detecting and preventing cyber insider threats: A survey. *IEEE Communications Surveys and Tutorials*, 20(2), 1397–1418. <https://doi.org/10.1109/COMST.2018.2800740>
- Makubhai, S., Pathak, G. R., & Chandre, P. R. (2023). *Prevention in healthcare : An explainable AI approach*. April (pp. 92–100).
- Maseno, E. M. (2017). *Vishing attack detection model for mobile users* (pp. 1–69). [pdf?sequence=1&isAllowed=y](https://doi.org/10.1109/ICCCI56745.2023.10128499).
- Molay, K. (2009). *White paper: Best practices for webinars*, 14.
- Opara, C., Chen, Y., & Wei, B. (2024). Look before you leap: Detecting phishing web pages by exploiting raw URL and HTML characteristics. *Expert Systems with Applications*, 236(October 2020), Article 121183. <https://doi.org/10.1016/j.eswa.2023.121183>
- Orunsolu, A. A., Sodiya, A. S., & Akinwale, A. T. (2022). A predictive model for phishing detection. *Journal of King Saud University - Computer and Information Sciences*, 34(2), 232–247. <https://doi.org/10.1016/j.jksuci.2019.12.005>
- Palaniappan, G., Sangeetha, S., Rajendran, B., Sanjay, Goyal, S., & Bindhumadhava, B. S. (2020). Malicious domain detection using machine learning on domain name features, host-based features and web-based features. *Procedia Computer Science*, 171(2019), 654–661. <https://doi.org/10.1016/j.procs.2020.04.071>
- Paliath, S., Qbeitah, M. A., & Aldwairi, M. (2020). Phishout: Effective phishing detection using selected features. In *Proceedings of the 2020 27th international conference on telecommunications, ICT 2020*. <https://doi.org/10.1109/ICT49546.2020.9239589>
- Punia, S. K., Kumar, M., Stephan, T., Deverajan, G. G., & Patan, R. (2021). Performance analysis of machine learning algorithms

- for big data classification: ML and ai-based algorithms for big data analysis. *International Journal of E-Health and Medical Communications*, 12(4), 60–75. <https://doi.org/10.4018/IJEHMC.20210701.oa4>
- Remmide, M. A., Boumahdi, F., Boustia, N., Feknous, C. L., & Della, R. (2022). Detection of phishing URLs using temporal convolutional network. *Procedia Computer Science*, 212(C), 74–82. <https://doi.org/10.1016/j.procs.2022.10.209>
- Ribeiro, L., Guedes, I. S., & Cardoso, C. S. (2024). Which factors predict susceptibility to phishing? An empirical study. *Computers & Security*, (October 2023), 136. <https://doi.org/10.1016/j.cose.2023.103558>
- Sánchez-Paniagua, M., Fidalgo, E., Alegre, E., & Alaiz-Rodríguez, R. (2022). Phishing websites detection using a novel multipurpose dataset and web technologies features. *Expert Systems with Applications*, 207(October 2021), Article 118010. <https://doi.org/10.1016/j.eswa.2022.118010>
- Sadiq, A., Anwar, M., Butt, R. A., Masud, F., Shahzad, M. K., Naseem, S., & Younas, M. (2021). A review of phishing attacks and countermeasures for internet of things-based smart business applications in industry 4.0. *Human Behavior and Emerging Technologies*, 3(5), 854–864. <https://doi.org/10.1002/hbe2.301>
- Safi, A., & Singh, S. (2023). A systematic literature review on phishing website detection techniques. *Journal of King Saud University - Computer and Information Sciences*, 35(2), 590–611. <https://doi.org/10.1016/j.jksuci.2023.01.004>
- Salloum, S., Gaber, T., Vadera, S., & Shaalan, K. (2022). A systematic literature review on phishing email detection using natural language processing techniques. *IEEE Access*, 10, 65703–65727. <https://doi.org/10.1109/ACCESS.2022.3183083>
- Sanchez-Paniagua, M., Fernandez, E. F., Alegre, E., Al-Nabki, W., & Gonzalez-Castro, V. (2022). Phishing URL detection: A real-case scenario through login URLs. *IEEE Access*, 10, 42949–42960. <https://doi.org/10.1109/ACCESS.2022.3168681>
- Sawe, L., Gikandi, J., Kamau, J., & Njuguna, D. (2024). Sentence level analysis model for phishing detection using KNN. *Journal of Cyber Security*, 6(0), 25–39. <https://doi.org/10.32604/jcs.2023.045859>
- Sutter, T., Bozkir, A. S., Gehring, B., & Berlich, P. (2022). Avoiding the hook: Influential factors of phishing awareness training on click-rates and a data-driven approach to predict email difficulty perception. *IEEE Access*, 10, 100540–100565. <https://doi.org/10.1109/ACCESS.2022.3207272>
- van Geest, R. J., Cascavilla, G., Hulstijn, J., & Zannone, N. (2024). The applicability of a hybrid framework for automated phishing detection. *Computers & Security*, 139(January), Article 103736. <https://doi.org/10.1016/j.cose.2024.103736>
- Williams, E. J., & Joinson, A. N. (2020). Developing a measure of information seeking about phishing. *Journal of Cybersecurity*, 6(1), 1–16. <https://doi.org/10.1093/cybsec/tyaa001>